Small-Signal Synchronization Stability of Sequence-Decomposed Grid-Forming IBRs

Qian Liu, Yanchang Liang, Zuan Zhang, Miao Wang,
and Xiaowei Zhao, *Member*, *IEEE*

***Abstract*—To facilitate relay operation and mitigate unbalanced voltage conditions, recent specifications mandate negative sequence control in grid-forming inverter-based resources (GFM IBRs). Adhering to this requirement, sequence-decomposed modules (SDMs) must be integrated into the control system to detect and extract sequence components. However, the introduction of SDMs induces cross-coupled effects in the *dq*-frame, potentially making GFM IBRs susceptible to sub-synchronous oscillations (SSOs)—an aspect not thoroughly investigated previously. This study addresses this gap by initially exploring the stability issue using single-input single-output small-signal models of GFM IBRs with SDMs. To mitigate SSO instability risks, a damping enhancement control scheme is then proposed, based on the fundamental hybrid synchronization control principle. By leveraging the *q*-axis positive sequence voltage from the SDM as a feedforward loop, system stability is significantly enhanced with promising damping capabilities. Simulation and controller hardware-in-the-loop tests, conducted across varying grid strengths, validate both the theoretical analysis and the effectiveness of the proposed control.**

***Index Terms*—Grid-forming control, inverter-based resources, small-signal stability, sequence-decomposed, negative sequence component.**

I. Introduction

THE global rise in penetration levels of inverter-based resources (IBRs), such as wind and solar photovoltaic, is projected to profoundly impact the dynamic performance of the power grid [1]. Unlike rotating electrical machines, which possess substantial mechanical inertia and naturally stabilize the grid by resisting rapid frequency changes, IBRs lack inherent inertia and can even exhibit zero inertia in some cases [2]. This discrepancy results in potential stability issues in modern power systems.

As rotating generators are increasingly phased out and more IBRs are integrated, it is crucial to evaluate inverter stability and develop effective control technology to ensure the stable operation of the power grid. A wide variety of instability issues have been reported for IBRs [3]-[10]. Transient stability concerns often arise from the limited thermal capacity of IBRs during large grid disturbances such as faults or sudden load changes [3]. The lack of damping in IBRs exacerbates these transient issues, leading to loss of synchronism problems. In transient stability studies, nonlinear large-scale models based on power angle, phase portrait, or energy function techniques [4], [5] are commonly developed for stability assessment, and adaptive control schemes are usually preferable for transient stability enhancement [6].

Due to the multiple timescales in IBR control systems, oscillations can be induced across a wide frequency range by interactions between IBRs and power grids [7]. In this context, numerous small-signal stability issues are studied via linearization methods around a certain equilibrium point, such as multi-input multi-output (MIMO) state-space or impedance modelling and intuitive single-input single-output (SISO) approaches [8], [9]. Once the specific root cause is identified, proper control design or improved outer and inner control loops can be implemented in IBRs to achieve robust small-signal stability [10].

Grid-forming (GFM) IBRs are known for their capabilities in regulating frequency and voltage magnitude in power grids. These IBRs have garnered significant attention in recent years due to their superior performance in providing inertia and maintaining stability under grid disturbances [11]. As an emerging technology in power systems, GFM IBRs are increasingly subject to regulatory frameworks [12]-[14] that mandate the provision of negative sequence currents under unbalanced grid conditions. The recently approved IEEE Standard 2800-2022 [15] stipulates all IBRs to generate negative sequence current in the presence of asymmetric faults. This requirement stems from the substantial impact that negative sequence currents or voltages from IBRs can have on a faulted network. Improper negative sequence components from IBRs can result in the mis-operation of negative sequence based protective elements [16] and cause overvoltage tripping of GFM IBRs [17]. Furthermore, several specifications [18], [19] even require GFM IBRs to provide negative sequence current during normal operation to maintain voltage balance. Thus, it is essential for GFM IBRs to incorporate negative sequence control in both normal and disturbed conditions.

Prior to implementing the negative sequence control for GFM IBRs, the negative sequence components should be first

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Qian Liu, Yanchang Liang, Miao Wang, and Xiaowei Zhao are with the Intelligent Control and Smart Energy (ICSE) Research Group, School of Engineering, University of Warwick, Coventry CV4 7AL, U.K. (e-mail: qian.liu.8@warwick.ac.uk; yanchang.liang@warwick.ac.uk; miao.wang.4@warwick.ac.uk; xiaowei.zhao@warwick.ac.uk).

Zuan Zhang is with the Siemens Energy, Manchester M20 2AF, U.K. (zuan.zhang@siemens-energy.com).

separated from the three-phase electrical quantities. This task is achieved through the utilization of the sequence-decomposed module (SDM) [20], [21]. SDMs are specifically designed to detect and extract the sequence components from real-world circuits, tailoring the voltage and current signals transmitted to the control system [22]. As a result, different feedforward and feedback loops are integrated to GFM IBRs by SDMs, thereby impacting the overall control loop dynamics and posing challenges to system stability.

Therefore, it is crucial to analyse the dynamic behaviours of different SDMs and unveil their effects on the operation of GFM IBRs. Previous studies have extensively investigated the small-signal modelling and synchronization stability of grid-following (GFL) IBRs with SDMs [23]-[25]. However, GFM IBRs distinguish themselves from GFL IBRs in their typical power-synchronized mechanism, rather than the phase-locked loop-synchronized approach [26]. Consequently, models and findings from GFL IBRs with SDMs cannot be directly extrapolated to their GFM counterparts.

Only a handful of research has delved into the small-signal stability analysis of GFM IBRs with SDMs. A recent study [27] derives an impedance model for GFM IBRs with SDMs in the stationary reference frame. This study includes the small-signal stability analysis and offers guidelines for parameter tuning, considering two typical SDMs. However, it is imperative to highlight that the synchronous reference frame, namely the *dq*-frame, is more commonly used in GFM IBRs [28]-[30]. Despite this, a noticeable gap remains, as there is currently no small-signal model for GFM IBRs with SDMs in the *dq*-frame. Moreover, to the best of our knowledge, no comparative stability analysis between GFM IBRs without and with SDMs has been reported to date. Additionally, an unresolved and critical issue concerns enhancing GFM control strategies to mitigate the impacts induced by SDMs.

To bridge the identified research gaps, this paper analyses the small-signal stability of GFM IBRs with the negative sequence control in the *dq*-frame. The sub-synchronous oscillation (SSO) instability issues arising from SDMs are explored using the developed SISO model. A control method is then proposed to improve the system small-signal stability. The main contributions are outlined as follows:

1) A generalized SISO small-signal model is established for GFM IBRs with various types of SDMs. On this basis, interactions between SDMs and the overall GFM control loops are characterized in detail.
2) Comparative small-signal stability analysis of GFM IBRs without and with different SDMs are presented, where the SSO issues inherent in GFM IBRs with SDMs are disclosed and comprehensively analysed.
3) A damping enhancement control (DAC) scheme based on the *q*-axis positive sequence voltage feedforward loop is proposed. This scheme effectively alleviates the above SSO issues across a wide range of AC grid strengths.

The remaining portions of the paper are organized as follows. Section II provides an overview of the GFM IBR with the SDM and its interconnected system. The linearized SISO modelling

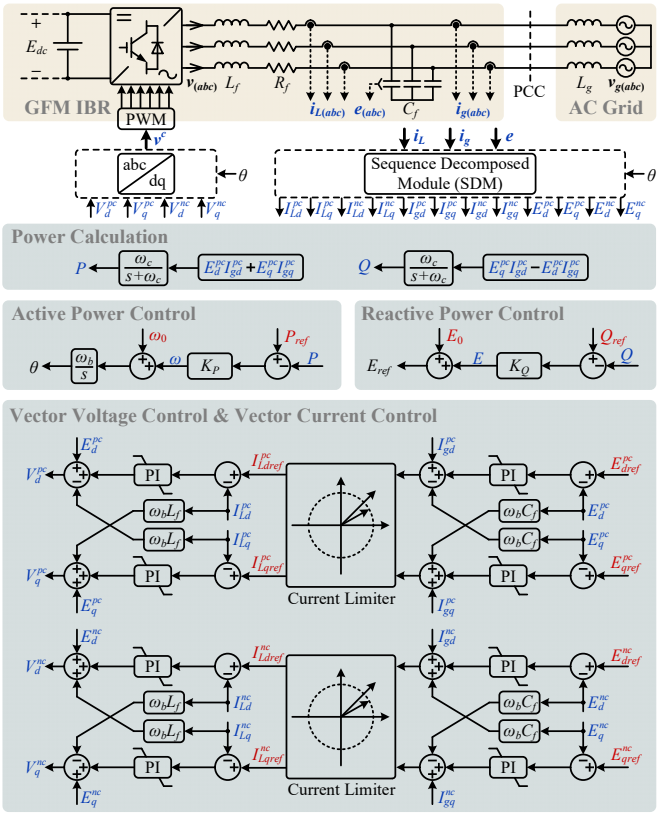

Fig. 1. System block diagram of the GFM IBR with SDMs.

TABLE I
PARAMETERS FOR SIMULATION AND CHIL TEST

Symbol	Description	Value
	GFM IBR	
S	Power rating	10 kVA
V_g	Rated voltage (L-L, RMS)	380 V
ω_b	Base angular frequency	100π rad/s
L_f	LC filter inductance	3 mH
R_f	LC filter resistance	0.1 Ω
C_f	LC filter capacitance	60 μF
ω_c	LPF cut-off frequency	10π rad/s
k_D	Constant used in DSOGI	1.414
T_D	Time constant used in DSC	5 ms
T_d	Control time delay	50 μs
	Droop Control	
ω_0	Nominal angular frequency	1 p.u.
E_0	Nominal PCC voltage	1 p.u.
P_{ref}	Active power reference	0.9 p.u.
Q_{ref}	Reactive power reference	0
K_P	APC droop gain	0.03 p.u.
K_Q	RPC droop gain	0.1 p.u.
Vector Control (bandwidth of VVC & VCC: 50 Hz & 500 Hz)		
k_{pv}	VVC proportional gain	0.273 p.u.
k_{iv}	VVC integral gain	0.855 p.u.
k_{pi}	VCC proportional gain	0.653 p.u.
k_{ii}	VCC integral gain	21.974 p.u.

procedures for GFM IBRs with SDMs are conducted in Section III, followed by a detailed small-signal stability analysis of the SSO issue, considering varied droop gains and AC grid conditions. In Section IV, the mechanism and parameter design guidelines of the proposed DAC scheme are expounded. Then, the theoretical analysis and the proposed method are validated through controller hardware-in-the-loop (CHIL) tests in Section V. Finally, Section VI summarizes the main findings.

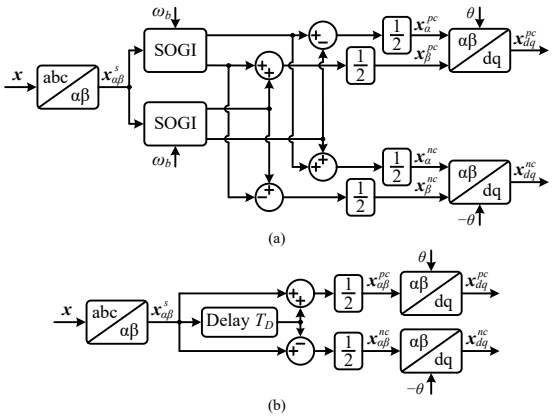

Fig. 2. Sequence-decomposed module. (a) DSOGI-based SDM. (b) DSC-based SDM. ($\boldsymbol{x}\in\{\boldsymbol{i_L}, \boldsymbol{i_g}, \boldsymbol{e}\}$)

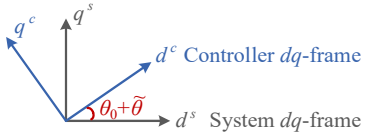

Fig. 3. AC grid and GFM IBR dq-frames.

II. System Description

A. System Configuration

As shown in Fig. 1, the studied system comprises a GFM IBR and the simplified AC grid, represented by a voltage source $\boldsymbol{v_g}$ in series with the inductance L_g (transformer inductance plus grid inductance) [31]. The GFM IBR is integrated to the point of common coupling (PCC) via an LC filter. The variables $\boldsymbol{v}$, $\boldsymbol{i_L}$, $\boldsymbol{i_g}$, and $\boldsymbol{e}$ correspond to the IBR terminal voltage, filter inductor current, grid current and PCC voltage in the stationary reference frame, respectively. Description of the circuit and control parameters in Fig. 1 can be found in TABLE I.

B. Sequence-Decomposed GFM Control System

To regulate both positive and negative sequence components in the dq-frame, signals $\boldsymbol{i_L}$, $\boldsymbol{i_g}$, and $\boldsymbol{e}$ must undergo processing by the SDM. This processing facilitates the generation of the positive and negative sequence signals that are then fed into the GFM control system.

There are five typical SDM structures [20]-[22], namely, dual synchronous reference frame (DSRF), dual complex-coefficient filter (DCCF), dual second-order generalized integrator (DSOGI), delayed signal cancellation (DSC), and notch filter (NF). Thereinto, DSRF, DCCF and DSOGI share the same mathematical expressions [20]. Besides, DSC and NF exhibit similar dynamics under the power frequency [22], falling within the frequency range of the SSO issues studied in this paper. For brevity, DSOGI- and DSC-based SDMs are considered for the following analysis, as presented in Fig. 2.

The active power control (APC) and reactive power control (RPC) adopt the typical droop control, rendering to

$$\omega = K_P\left(P_{ref} - P\right),\ \theta = \frac{\omega_b\left(\omega+\omega_0\right)}{s} \tag{1}$$

$$E = K_Q\left(Q_{ref} - Q\right) \tag{2}$$

where ω is the angular frequency error derived from the APC, θ is the output phase angle of the APC, E is the magnitude error of the PCC voltage, P and Q are the active and reactive power calculated using the power calculation module in Fig. 1.

The inner control loops consist of the commonly used vector voltage control (VVC) loop and vector current control (VCC) loop [28]-[30], and can be given by

$$\boldsymbol{I}^c_{Ldqref} = G_{VVC}\left(\boldsymbol{E}^c_{dqref} - \boldsymbol{E}^c_{dq}\right) + \boldsymbol{J}\omega_b C_f \boldsymbol{E}^c_{dq} + \boldsymbol{I}^c_{gdq} \tag{3}$$

$$\boldsymbol{V}^c_{dq} = G_{VCC}\left(\boldsymbol{I}^c_{Ldqref} - \boldsymbol{I}^c_{Ldq}\right) + \boldsymbol{J}\omega_b L_f \boldsymbol{I}^c_{Ldq} + \boldsymbol{E}^c_{dq} \tag{4}$$

where the boldface letters can be expressed as $\boldsymbol{I}^c_{Ldqref}$= $[I^{pc}_{Ldref}, I^{pc}_{Lqref}, I^{nc}_{Ldref}, I^{nc}_{Lqref}]^T$, $\boldsymbol{E}^c_{dqref}$= $[E^{pc}_{dref}, E^{pc}_{qref}, E^{nc}_{dref}, E^{nc}_{qref}]^T$, $\boldsymbol{E}^c_{dq}$= $[E^{pc}_{d}, E^{pc}_{q}, E^{nc}_{d}, E^{nc}_{q}]^T$, $\boldsymbol{V}^c_{dq}$= $[V^{pc}_{d}, V^{pc}_{q}, V^{nc}_{d}, V^{nc}_{q}]^T$, $\boldsymbol{I}^c_{gdq}$= $[I^{pc}_{gd}, I^{pc}_{gq}, I^{nc}_{gd}, I^{nc}_{gq}]^T$, $\boldsymbol{I}^c_{Ldq}$= $[I^{pc}_{Ld}, I^{pc}_{Lq}, I^{nc}_{Ld}, I^{nc}_{Lq}]^T$, G_{VVC}= $(k_{pv}+k_{iv}/s)$, G_{VCC}= $(k_{pi}+k_{ii}/s)$. Superscript c indicates variables in the GFM IBR reference dq-frame (hereafter referred to as controller dq-frame), superscripts p and n denote the positive and negative sequence components, respectively, and superscript T indicates the matrix transpose. Subscripts d and q represent the d- and q-axis variables, and

$$\boldsymbol{J} = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix} \tag{5}$$

Through rotation transformations, the modulation voltage vector $\boldsymbol{v^c}$ can be generated. The control time delay is expressed as $\boldsymbol{G}_d = \boldsymbol{I}_{44}e^{-sT_d}$, where $\boldsymbol{I}_{44}$ refers to the 4×4 identity matrix.

III. SSO Issues of GFM IBR with SDMs

To elucidate the adverse stability effect of SDMs, the system in Fig. 1 is simplified from a MIMO model to a SISO model (from P_{ref} to P). This model is then used to perform the small-signal stability analysis.

A. SISO Small-Signal Model of GFM IBR with SDMs

Recalling (1), and assuming a small-signal perturbation in the APC, the phase angle dynamics are depicted as

$$\tilde{\theta} = \frac{\omega_b}{s}K_P\left(\tilde{P}_{ref} - \tilde{P}\right) \tag{6}$$

where variables with symbol ~ are their small-signal forms.

The impact of $\tilde{\theta}$ determines the phase angle relationship between the AC grid reference dq-frame (hereafter referred to as system dq-frame) and the controller dq-frame, as presented in Fig. 3, where θ_0 is the initial phase angle difference, and superscript s indicates variables in the system dq-frame.

In accordance with (6) and Fig. 3, the system closed-loop small-signal dynamics can be developed by considering both the control loop dynamics and circuit dynamics.

1) SDM: Given that the input variables of different control loops in Fig. 1 are all derived from the SDM, mathematical expressions of the SDM need to be firstly explored.

Based on Fig. 2(a), the characteristic transfer function of the DSOGI-based SDM can be written in the $\alpha\beta$-frame as [20]

$$\boldsymbol{X}_{\alpha\beta}^{c}=\boldsymbol{K}\boldsymbol{X}_{\alpha\beta}^{s}=\frac{1}{2}\frac{k_D\omega_b}{s^2+k_D\omega_b s+\omega_b{}^2}\begin{bmatrix} s & -\omega_b \\ \omega_b & s \\ s & \omega_b \\ -\omega_b & s \end{bmatrix}\boldsymbol{X}_{\alpha\beta}^{s} \quad (7)$$

where $\boldsymbol{X}_{\alpha\beta}^{c}=[X_{\alpha}^{pc}, X_{\beta}^{pc}, X_{\alpha}^{nc}, X_{\beta}^{nc}]^T$, $\boldsymbol{X}_{\alpha\beta}^{s}=[X_{\alpha}^{s}, X_{\beta}^{s}]^T$, and $X\in\{I_L, I_g, E\}$.

To obtain time domain variables in the controller dq-frame, the 4×4 rotation transformation matrix $\boldsymbol{T}(\theta)$ is defined as

$$\boldsymbol{T}(\theta)=\left[\begin{array}{cc:cc} \cos(\theta) & \sin(\theta) & 0 & 0 \\ -\sin(\theta) & \cos(\theta) & 0 & 0 \\ \hdashline 0 & 0 & \cos(\theta) & -\sin(\theta) \\ 0 & 0 & \sin(\theta) & \cos(\theta) \end{array}\right] \quad (8)$$

Applying (8) to both sides of (7) in the time domain yields

$$\begin{aligned}\boldsymbol{x}_{dq}^{c}&=\boldsymbol{T}(\theta)\left\{\mathcal{L}^{-1}[\boldsymbol{K}]*\left[\boldsymbol{C}\boldsymbol{T}^{-1}(\omega_b t)\boldsymbol{x}_{dq}^{s}\right]\right\}\\&=\boldsymbol{T}(\omega_b t+\theta_0)\boldsymbol{z}_{DSOGI}\end{aligned} \quad (9)$$

where the lowercase bold letters represents the time domain variables, $\boldsymbol{x}_{dq}^{c}=[x_d^{pc}, x_q^{pc}, x_d^{nc}, x_q^{nc}]^T$, $\boldsymbol{x}_{dq}^{s}=[x_d^{ps}, x_q^{ps}, x_d^{ns}, x_q^{ns}]^T$, operator * denotes the convolution operation, and

$$\boldsymbol{C}=\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \quad (10)$$

The small-signal expression of (9) is considered in the time domain as

$$\tilde{\boldsymbol{x}}_{dq}^{c}=\boldsymbol{T}(\omega_b t+\theta_0)\tilde{\boldsymbol{z}}_{DSOGI}+\boldsymbol{G}_X^{\theta}\tilde{\theta} \quad (11)$$

where $\boldsymbol{G}_X^{\theta}=-\boldsymbol{J}\boldsymbol{X}_{dq0}^{c}$, $\boldsymbol{X}_{dq0}^{c}$ is the steady-state operating point of $\boldsymbol{X}_{dq}^{c}$, $\boldsymbol{X}_{dq0}^{c}=[X_{d0}^{pc}, X_{q0}^{pc}, X_{d0}^{nc}, X_{q0}^{nc}]^T$, and variables with subscript 0 denote their values at the steady-state operating point.

By taking the Laplace transform on (11), the small-signal dynamics of the DSOGI-based SDM can be determined as

$$\tilde{\boldsymbol{X}}_{dq}^{c}=\boldsymbol{T}(\theta_0)\boldsymbol{H}^{DSOGI}\tilde{\boldsymbol{X}}_{dq}^{s}+\boldsymbol{G}_X^{\theta}\tilde{\theta} \quad (12)$$

where

$$\boldsymbol{H}^{DSOGI}=\frac{k_D\omega_b}{2H_f}\left[\begin{array}{cc:cc} H_1^{DSOGI} & H_2^{DSOGI} & 0 & 0 \\ -H_2^{DSOGI} & H_1^{DSOGI} & 0 & 0 \\ \hdashline 0 & 0 & H_1^{\overline{DSOGI}} & -H_2^{\overline{DSOGI}} \\ 0 & 0 & H_2^{DSOGI} & H_1^{DSOGI} \end{array}\right]$$

$$\begin{aligned}H_1^{DSOGI}&=s^3+k_D\omega_b s^2+4\omega_b^2 s+2k_D\omega_b^3\\H_2^{DSOGI}&=-k_D\omega_b^2 s\\H_f^{DSOGI}&=s^4+2k_D\omega_b s^3+\left(k_D^2+4\right)\omega_b^2 s^2+4k_D\omega_b^3 s+k_D^2\omega_b^4\end{aligned} \quad (13)$$

Similarly, the transfer function of DSC-based SDM in the $\alpha\beta$-frame can be obtained by referring to Fig. 2(b), and manipulated using the mathematical derivations in (8)–(11). By incorporating the first-order Padé approximation [22], [23], small-signal dynamics of the DSC-based SDM are yielded as

$$\tilde{\boldsymbol{X}}_{dq}^{c}=\boldsymbol{T}(\theta_0)\boldsymbol{H}^{DSC}\tilde{\boldsymbol{X}}_{dq}^{s}+\boldsymbol{G}_X^{\theta}\tilde{\theta} \quad (14)$$

where

$$\boldsymbol{H}^{DSC}=\left[\begin{array}{cc:cc} H^{DSC} & 0 & 0 & 0 \\ 0 & H^{DSC} & 0 & 0 \\ \hdashline 0 & 0 & H^{DSC} & 0 \\ 0 & 0 & 0 & H^{DSC} \end{array}\right] \quad (15)$$

$$H^{DSC}=1/(T_D s/2+1)$$

Based on (12) and (14), the small-signal expressions of the SDM can be summarized below

$$\tilde{\boldsymbol{X}}_{dq}^{c}=\boldsymbol{H}\tilde{\boldsymbol{X}}_{dq}^{s}+\boldsymbol{G}_X^{\theta}\tilde{\theta} \quad (16)$$

where $\boldsymbol{H}=\boldsymbol{T}(\theta_0)\boldsymbol{H}^{DSOGI}$ for the DSOGI-based SDM, and $\boldsymbol{H}=\boldsymbol{T}(\theta_0)\boldsymbol{H}^{DSC}$ for the DSC-based SDM.

Note that the frequency coupling components are omitted in (12)–(15) [25], since they fall outside the frequency range of interest in this study.

From (7) to (16), it is apparent that the inclusion of the DSOGI-based SDM introduces cross couplings in the d-axis and q-axis components of $\tilde{\boldsymbol{X}}_{dq}^{c}$ through the matrix $\boldsymbol{H}^{DSOGI}$. Although the DSC-based SDM does not result in coupling properties, it imports low-pass characteristics in the control variables, inducting control delay to the GFM control system. Additionally, it can be seen that $\tilde{\boldsymbol{X}}_{dq}^{c}$ interacts with the APC loop via $\tilde{\theta}$. Note that $\tilde{\boldsymbol{X}}_{dq}^{c}$ serves as the input to various control loops, as illustrated in Fig. 1. Consequently, the above coupling, delay, and interaction effects complicate the control dynamics of the sequence-decomposed GFM IBR, potentially compromising system stability.

2) Power Calculation: As presented in Fig. 1, P and Q are calculated using the positive sequence components of $\boldsymbol{E}^{c}$ and $\boldsymbol{I}_g^{c}$, and their small-signal dynamics are expressed as

$$\tilde{P}=F_p\left(\boldsymbol{G}_P^{e}\boldsymbol{I}_{24}\tilde{\boldsymbol{E}}_{dq}^{c}+\boldsymbol{G}_P^{i}\boldsymbol{I}_{24}\tilde{\boldsymbol{I}}_{gdq}^{c}\right) \quad (17)$$

$$\tilde{Q}=F_p\left(\boldsymbol{G}_Q^{e}\boldsymbol{I}_{24}\tilde{\boldsymbol{E}}_{dq}^{c}+\boldsymbol{G}_Q^{i}\boldsymbol{I}_{24}\tilde{\boldsymbol{I}}_{gdq}^{c}\right) \quad (18)$$

where $\boldsymbol{G}_P^{e}=[I_{gd0}^{pc}, I_{gq0}^{pc}]$, $\boldsymbol{G}_P^{i}=[E_{d0}^{pc}, E_{q0}^{pc}]$, $\boldsymbol{G}_Q^{e}=[-I_{gq0}^{pc}, I_{gd0}^{pc}]$, $\boldsymbol{G}_Q^{i}=[E_{q0}^{pc}, -E_{d0}^{pc}]$, $F_p=\omega_c/(s+\omega_c)$ denotes the low-pass filter (LPF) used in power calculation, and matrix $\boldsymbol{I}_{24}$ refers to

$$\boldsymbol{I}_{24}=\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (19)$$

3) APC and RPC: Small-signal dynamics of the APC are determined by (6), while those of the RPC are given by

$$\tilde{E}_{ref}=K_Q\left(\tilde{Q}_{ref}-\tilde{Q}\right) \quad (20)$$

The positive and negative sequence voltage references can be calculated as

$$\tilde{\boldsymbol{E}}_{dqref}^{c}=\begin{bmatrix}1 & 0 & 0 & 0\end{bmatrix}^T\tilde{E}_{ref}=\boldsymbol{I}_{41}\tilde{E}_{ref} \quad (21)$$

4) VVC and VCC: Based on (3) and (4), the equivalent small-signal functions of VVC and VCC are

$$\tilde{\boldsymbol{I}}_{Ldqref}^{c}=G_{VVC}\left(\tilde{\boldsymbol{E}}_{dqref}^{c}-\tilde{\boldsymbol{E}}_{dq}^{c}\right)+\boldsymbol{J}\omega_b C_f\tilde{\boldsymbol{E}}_{dq}^{c}+\tilde{\boldsymbol{I}}_{gdq}^{c} \quad (22)$$

$$\tilde{\boldsymbol{V}}_{dq}^{c}=G_{VCC}\left(\tilde{\boldsymbol{I}}_{Ldqref}^{c}-\tilde{\boldsymbol{I}}_{Ldq}^{c}\right)+\boldsymbol{J}\omega_b L_f\tilde{\boldsymbol{I}}_{Ldq}^{c}+\tilde{\boldsymbol{E}}_{dq}^{c} \quad (23)$$

The preceding derivation results mainly pertain to the control loop dynamics. Then, the circuit dynamics will be conducted to

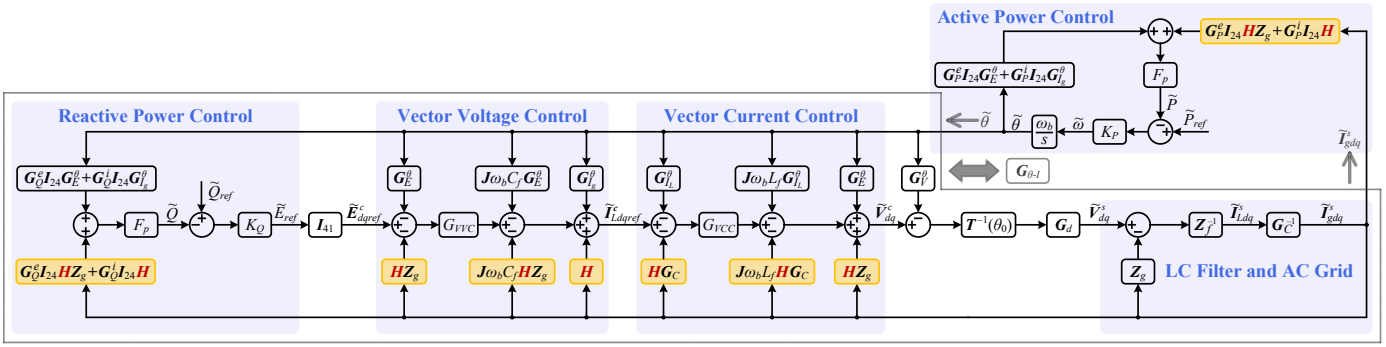

Fig. 4. Block diagram of the system small-signal model considering the GFM IBR with SDMs.

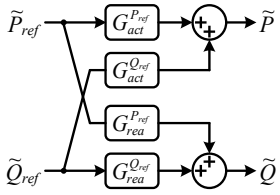

Fig. 5. Block diagram of the MIMO system.

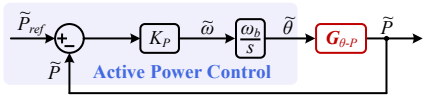

Fig. 6. Block diagram of the simplified SISO system.

establish the overall closed-loop system dynamics.

5) AC Grid Dynamics: The AC grid dynamics can be directly described in the system *dq*-frame as

$$\tilde{\boldsymbol{V}}^s_{gdq} = \tilde{\boldsymbol{E}}^s_{dq} - \boldsymbol{Z}_g \tilde{\boldsymbol{I}}^s_{gdq} = 0 \Rightarrow \tilde{\boldsymbol{E}}^s_{dq} = \boldsymbol{Z}_g \tilde{\boldsymbol{I}}^s_{gdq} \tag{24}$$

where $\tilde{\boldsymbol{V}}^s_{gdq} = [\tilde{V}^{ps}_{gd}, \tilde{V}^{ps}_{gq}, \tilde{V}^{ns}_{gd}, \tilde{V}^{ns}_{gq}]^T$, $\tilde{\boldsymbol{E}}^s_{dq} = [\tilde{E}^{ps}_{d}, \tilde{E}^{ps}_{q}, \tilde{E}^{ns}_{d}, \tilde{E}^{ns}_{q}]^T$, $\tilde{\boldsymbol{I}}^s_{gdq} = [\tilde{I}^{ps}_{gd}, \tilde{I}^{ps}_{gq}, \tilde{I}^{ns}_{gd}, \tilde{I}^{ns}_{gq}]^T$, and the impedance matrix $\boldsymbol{Z}_g$ is

$$\boldsymbol{Z}_g = \left[\begin{array}{cc|cc} sL_g & -\omega_b L_g & 0 & 0 \\ \omega_b L_g & sL_g & 0 & 0 \\ \hline 0 & 0 & sL_g & \omega_b L_g \\ 0 & 0 & -\omega_b L_g & sL_g \end{array}\right] \tag{25}$$

6) LC Filter Dynamics: The small-signal dynamics of the LC filter comprise two separate parts. First, relations between the filter current and grid current can be written as

$$\tilde{\boldsymbol{I}}^s_{Ldq} = \tilde{\boldsymbol{I}}^s_{gdq} + \boldsymbol{Y}_C \tilde{\boldsymbol{E}}^s_{dq} = \boldsymbol{G}_C \tilde{\boldsymbol{I}}^s_{gdq} \tag{26}$$

where $\tilde{\boldsymbol{I}}^s_{Ldq} = [\tilde{I}^{ps}_{Ld}, \tilde{I}^{ps}_{Lq}, \tilde{I}^{ns}_{Ld}, \tilde{I}^{ns}_{Lq}]^T$, $\boldsymbol{G}_C = \boldsymbol{I}_{44} + \boldsymbol{Y}_C \boldsymbol{Z}_g$, and $\boldsymbol{Y}_C$ can be similarly defined based on $\boldsymbol{Z}_g$.

Next, interactions between the IBR terminal voltage and PCC voltage are considered as

$$\tilde{\boldsymbol{V}}^s_{dq} = \tilde{\boldsymbol{E}}^s_{dq} + \boldsymbol{Z}_f \tilde{\boldsymbol{I}}^s_{Ldq} \tag{27}$$

where $\tilde{\boldsymbol{V}}^s_{dq} = [\tilde{V}^{ps}_{d}, \tilde{V}^{ps}_{q}, \tilde{V}^{ns}_{d}, \tilde{V}^{ns}_{q}]^T$, and $\boldsymbol{Z}_f$ can be analogously determined by (25).

7) Closed-Loop System Dynamics: To integrate the above expressions with *1)* through *6)*, an essential consideration is the mathematical relationship between the modulation voltage and IBR terminal voltage, which can be given by

$$\boldsymbol{G}_d \boldsymbol{T}^{-1}(\theta_0)\left(\tilde{\boldsymbol{V}}^c_{dq} - \boldsymbol{G}^{\theta}_V \tilde{\theta}\right) = \tilde{\boldsymbol{V}}^s_{dq} \tag{28}$$

where $\boldsymbol{G}^{\theta}_V = \boldsymbol{T}(\theta_0)\boldsymbol{V}^c_{dq0}$, and $\boldsymbol{V}^c_{dq0} = [V^{pc}_{q0}, -V^{pc}_{d0}, -V^{nc}_{q0}, V^{nc}_{d0}]^T$.

Referencing the known information in (16)–(28), and after some simple mathematical manipulations, the small-signal model for the GFM IBR with SDMs is presented in Fig. 4.

The main impacts of the SDM on the small-signal model are indicated with yellow boxes in Fig. 4, where $\boldsymbol{H}$ is highlighted in red to illustrate the dynamics induced by SDMs. Observation of Fig. 4 reveals that the SDM influences all control loops, including APC, RPC, VVC, and VCC. Moreover, by substituting $\boldsymbol{H}$ with the identity matrix $\boldsymbol{I}_{44}$ and removing all negative sequence components, the proposed model in Fig. 4 can also be represented as the small-signal model for the GFM IBR without SDMs.

Fig. 4 shows a typical MIMO system with two inputs $\tilde{P}_{ref}$ and $\tilde{Q}_{ref}$. By opting variables $\tilde{P}$ and $\tilde{Q}$ as outputs and performing mathematical manipulations on each loop in Fig. 4, the MIMO system is rearranged as shown in Fig. 5. In this configuration, the four individual SISO closed-loop transfer functions $G^{P_{ref}}_{act}$, $G^{Q_{ref}}_{act}$, $G^{P_{ref}}_{rea}$, and $G^{Q_{ref}}_{rea}$ can be used to evaluate the overall system small-signal stability. Notably, these transfer functions share the same polynomials in the denominator when expressed using the elements in Fig. 4. Consequently, a single SISO stability analysis is sufficient to assess the small-signal stability of the entire MIMO system in Fig. 5, encompassing both *d*- and *q*-axes. This paper then assumes $\tilde{Q}_{ref} = 0$ for two reasons: 1) The overall system stability can be accurately and efficiently evaluated using a single SISO transfer function $G^{P_{ref}}_{act}$. 2) The step response of the active power can be easily calculated using $G^{P_{ref}}_{act}$. Additionally, the stability margin can be intuitively identified through the SISO system from $\tilde{P}_{ref}$ to $\tilde{P}$. By combining the step response and stability margin, effective control design can be achieved, considering both time and frequency domain dynamics.

It is worth mentioning that the SISO analysis is difficult to directly apply to a system with more than two oscillation areas due to the complexity in obtaining the grid impedance. By assuming $\tilde{P}_{ref} = 0$ and $\tilde{Q}_{ref} = 0$, and removing the grid impedance in Fig. 4, the model can be transformed into a MIMO output-impedance (from $\tilde{\boldsymbol{I}}^s_{gdq}$ to $\tilde{\boldsymbol{E}}^s_{dq}$) or a MIMO input-admittance model (from $\tilde{\boldsymbol{E}}^s_{dq}$ to $\tilde{\boldsymbol{I}}^s_{gdq}$). With the impedance or admittance model, each GFM IBR behaves as a passive system, allowing the application of a stability criterion to evaluate the stability of

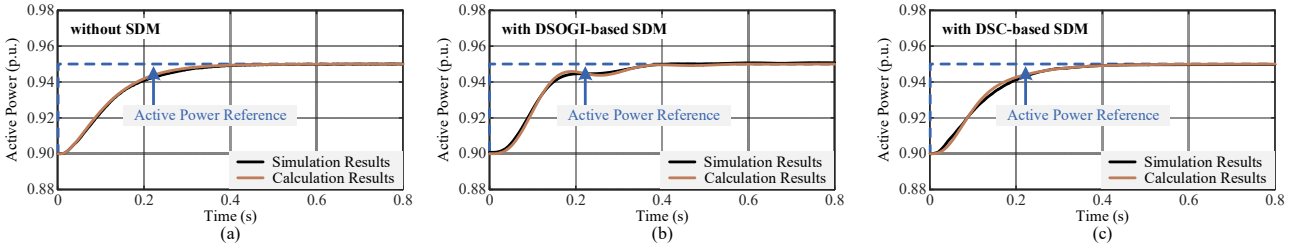

Fig. 7. Step response of the active power. (a) GFM IBR without SDMs. (b) GFM IBR with the DSOGI-based SDM. (c) GFM IBR with the DSC-based SDM.

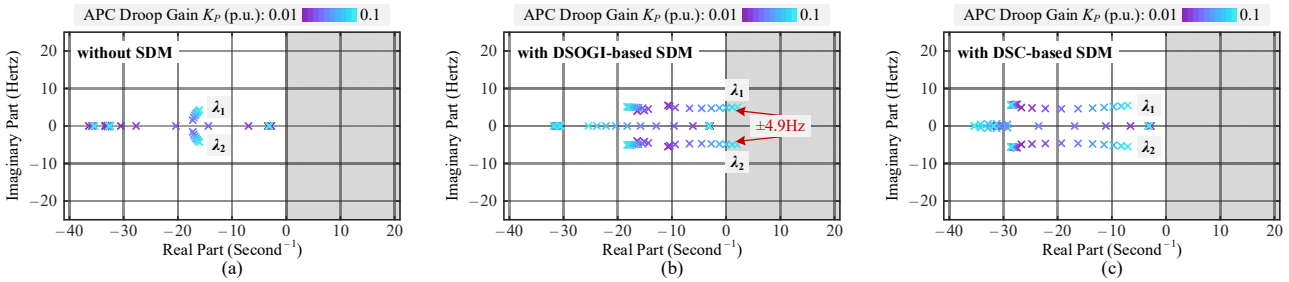

Fig. 8. Closed-loop pole trajectories of the SISO model under weak grid conditions (L_g= 0.75 p.u., SCR= 1.3). (a) GFM IBR without SDMs. (b) GFM IBR with the DSOGI-based SDM. (c) GFM IBR with the DSC-based SDM.

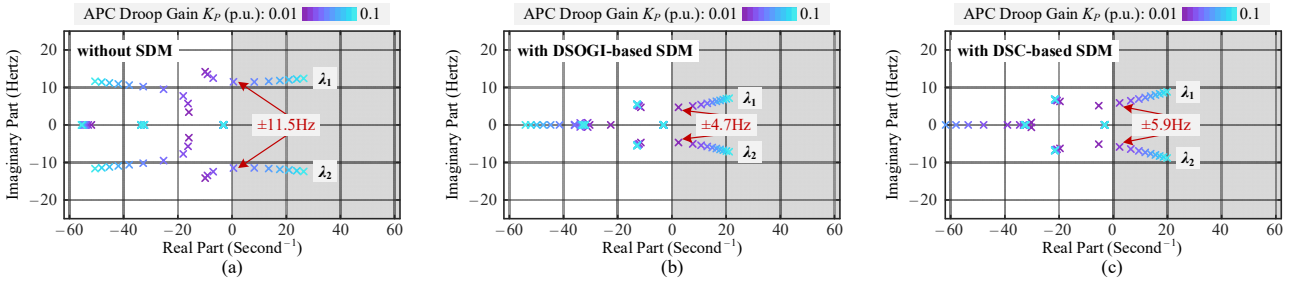

Fig. 9. Closed-loop pole trajectories of the SISO model under strong grid conditions (L_g= 0.15 p.u., SCR= 6.7). (a) GFM IBR without SDMs. (b) GFM IBR with the DSOGI-based SDM. (c) GFM IBR with the DSC-based SDM.

a multi-area system. This demonstrates the favorable scalability of the developed model.

With $\widetilde{Q}_{ref}$= 0, the dynamics depicted by the gray solid lines in Fig. 4 can be derived to obtain the mathematical relationship between grid current and phase angle, denoted as $\boldsymbol{G}_{\theta\text{-}I}$. Furthermore, the APC in Fig. 4 shows that $\widetilde{P}$ can be rewritten using $\tilde{\theta}$ and $\tilde{\boldsymbol{I}}^{s}_{gdq}$ as

$$\tilde{P} = F_p \begin{bmatrix} \left(\boldsymbol{G}^{e}_{P}\boldsymbol{I}_{24}\boldsymbol{G}^{\theta}_{E} + \boldsymbol{G}^{i}_{P}\boldsymbol{I}_{24}\boldsymbol{G}^{\theta}_{I_g}\right)\tilde{\theta} + \\ \left(\boldsymbol{G}^{e}_{P}\boldsymbol{I}_{24}\boldsymbol{H}\boldsymbol{Z}_{g} + \boldsymbol{G}^{i}_{P}\boldsymbol{I}_{24}\boldsymbol{H}\right)\tilde{\boldsymbol{I}}^{s}_{gdq} \end{bmatrix} \tag{29}$$

By substituting $\tilde{\boldsymbol{I}}^{s}_{gdq}$= $\boldsymbol{G}_{\theta\text{-}I}\tilde{\theta}$ to (29), $\widetilde{P}$ can be expressed as $\widetilde{P}$= $G_{\theta\text{-}P}\tilde{\theta}$. Given $G_{\theta\text{-}P}$, the small-signal model can be simplified to a straightforward unit negative feedback system, as shown in Fig. 6. Thus, the open-loop small-signal transfer function TF_{ol} and its closed-loop counterpart TF_{cl} can be determined as

$$TF_{ol} = \frac{G_{\theta\text{-}P}\omega_b K_P}{s}, \quad TF_{cl} = \frac{TF_{ol}}{1+TF_{ol}} \tag{30}$$

It is necessary to validate the accuracy of the developed model before conducting the small-signal stability analysis. Fig. 7 shows the step response of the active power from the developed model and the time-domain simulation model established in PSCAD/EMTDC, where the used parameters are given in TABLE I and L_g= 0.75 p.u. (SCR= 1.3) is taken as an example. The results from the derived SISO model and the simulation closely overlap in Fig. 7, confirming the accuracy of the proposed model.

B. Analysis of SSO Issues Induced by SDMs

Based on the proposed SISO model, the closed-loop pole trajectories can be calculated using TF_{cl} in (30) to evaluate the small-signal stability of the GFM IBR under different grid conditions. Fig. 8 and Fig. 9 show the pole trajectory results in weak (SCR= 1.3) and strong (SCR= 6.7) grid conditions, respectively. The APC droop gain K_P is varied from 0.01 p.u. to 0.1 p.u. [11], [29], [30], while other parameters are detailed in TABLE I. The pole loci near the imaginary axis are presented in Fig. 8 and Fig. 9, where a pair of sub-synchronous frequency conjugate poles labeled as "λ_1" and "λ_2" (referred to as SSO mode conjugate poles) dominate system stability.

1) Weak Grid Condition: From Fig. 8(a), it can be observed that the damping ratio of the SSO mode slightly decreases as K_P increases. Notably, the adjustable range of K_P covers a wide span from 0.01 p.u. to 0.1 p.u., implying favorable stability characteristics of the GFM IBR without SDMs under weak grid conditions. However, in Fig. 8(b), the SSO mode conjugate poles $\lambda_{1\text{-}2}$ are in the right-half plane (RHP) when K_P exceeds 0.08 p.u., demonstrating the negative impact of the DSOGI-based SDM on system stability. With a limited range of optional

K_P values, the GFM IBR using DSOGI-based SDM becomes vulnerable to SSO issues in a weak grid. Likewise, the inclusion of the DSC-based SDM in the GFM IBR induces a significant pole shift towards the RHP, as shown in Fig. 8(c). Despite the declined damping ratio of the SSO mode, the GFM IBR employing the DSC-based SDM presents a superior stability boundary compared to that with the DSOGI-based SDM, capable of withstanding K_P values up to 0.1 p.u.

2) Strong Grid Condition: Due to the low output impedance attributed to VVC, the GFM IBR exhibits a poor stability under the strong grid condition [32]. In Fig. 9(a), it is evident that the GFM IBR has a narrow range of available K_P values even without SDMs, specifically ranging from 0.01 p.u. to 0.03 p.u. The introduction of SDMs further deteriorates the stability, as illustrated in Fig. 9(b)(c). The results indicate that the GFM IBR with the DSOGI-based SDM performs inferior damping capability compared to its DSC-based SDM counterpart, and the former even requires a K_P value lower than 0.01 p.u. to ensure a steady-state operating point of the system. As a result, it is challenging for GFM IBRs with SDMs to stably operate in a stiff grid. By comparing the pole loci in Fig. 9(a) with those in Fig. 9(b)(c), it is also noticeable that the oscillation frequency of the unstable poles are diminished by SDMs, decreasing from 11.5 Hz to below 10 Hz.

3) Brief Summary: Despite the advantages of SDMs in facilitating the sequence-decomposed GFM control, they pose a detrimental impact on system stability of the GFM IBR in both weak and strong grid conditions. Specifically, the SSO mode conjugate poles λ_1 and λ_2 can easily shift to the RHP under the influence of SDMs, particularly with the DSOGI-based SDM. Referring to the proposed model, the high-order and coupling characteristics of the DSOGI-based SDM in (12) and (13) directly impact the APC, RPC, VVC, and VCC via the matrix $\boldsymbol{H}$ in Fig. 4, while the matrix $\boldsymbol{H}$ induced by the DSC-based SDM in (14) and (15) mainly imposes low-pass features on the control dynamics. Among all the control loops, the droop gain of the APC dominates system stability, because K_P directly determines the synchronization dynamics between the sequence-decomposed GFM IBR and AC grid. Other control parameters, such as the droop gain of RPC, control bandwidth of VVC, and control bandwidth of VCC have marginal impacts on the synchronization stability due to their low correlation or high bandwidth properties to the synchronism. By introducing SDMs to the GFM control system, the frequency response of the interconnection system is significantly reshaped, resulting in unstable SSO issues.

IV. Proposed Damping Enhancement Control

The analysis in Section III highlights that the GFM IBR with SDMs is always prone to the SSO issue across a wide range of grid conditions. To address this issue, a DAC is proposed to mitigate the adverse stability impacts of SDMs.

A. Basic Principle

Inspired by the general principle of hybrid synchronization control (HSC) [33], [34], the q-axis voltage signal incorporates phase angle dynamics akin to those of the active power. This attribute renders the q-axis voltage well-suited for damping purposes. Therefore, the q-axis positive sequence voltage E_q^{pc} is leveraged to enhance the damping capability of the GFM IBR with SDMs, as presented in Fig. 10. The proposed DAC operates in parallel with the APC and uses a proportional-integral controller G_E= (K_{pE}+K_{iE}/s), where K_{pE} and K_{iE} denote the proportional gain and integral gain, respectively.

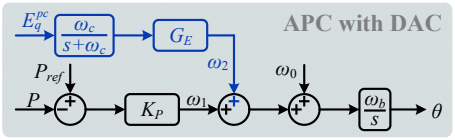

Fig. 10. Proposed damping enhancement control based on the q-axis positive sequence voltage feedforward.

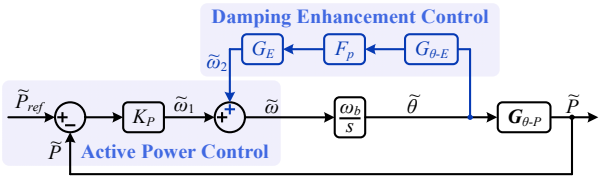

Fig. 11. SISO small-signal model of GFM IBR with SDMs considering the proposed damping enhancement control.

The difference between the HSC and the proposed DAC lies in the q-axis voltage extraction method employed in the control system. In contrast to directly transforming the q-axis voltage signal using the PCC voltage $\boldsymbol{e}$, the proposed DAC exploits the q-axis positive sequence voltage signal E_q^{pc} processed by the SDM. This modification makes the proposed DAC particularly suitable for GFM IBRs with SDMs.

B. Small-Signal Stability Analysis with DAC

Prior to conducting the stability analysis, it is essential to develop the small-signal model of the GFM IBR with DAC. Based on (16), (24), and $\boldsymbol{G}_{\theta\text{-}I}$, the small-signal expression of the input signal E_q^{pc} in Fig. 10 can be expressed as

$$\tilde{E}_q^{pc} = \boldsymbol{I}_{14}\left(\boldsymbol{H}\boldsymbol{Z}_g\tilde{\boldsymbol{I}}_{gdq}^{s} + \boldsymbol{G}_E^{\theta}\tilde{\theta}\right) = \underbrace{\boldsymbol{I}_{14}\left(\boldsymbol{H}\boldsymbol{Z}_g\boldsymbol{G}_{\theta\text{-}I} + \boldsymbol{G}_E^{\theta}\right)}_{G_{\theta\text{-}E}}\tilde{\theta} \quad (31)$$

where $\boldsymbol{I}_{14}$= [0, 1, 0, 0].

By combining Fig. 6, Fig. 10, and further using (31), the SISO small-signal model with DAC taken into account is illustrated in Fig. 11. The open-loop and closed-loop transfer functions can then be derived from Fig. 11 as

$$TF_{ol} = \frac{G_{\theta\text{-}P}\omega_b K_P}{s - \omega_b G_E F_p G_{\theta\text{-}E}}, \quad TF_{cl} = \frac{TF_{ol}}{1 + TF_{ol}} \quad (32)$$

The damping enhancement mechanism underlying the proposed control method is illustrated as follows.

Based on the system topology described in Fig. 1, the positive sequence PCC voltage can be expressed as

$$E_q^{ps} = \omega_b L_g I_{gd}^{ps} + V_{gq}^{ps} \quad (33)$$

Assuming the power angle between the PCC voltage phasor and the grid voltage phasor is δ= θ–θ_g, where θ_g represents the phase angle of the grid voltage, the q-axis positive sequence grid voltage can then be rewritten as

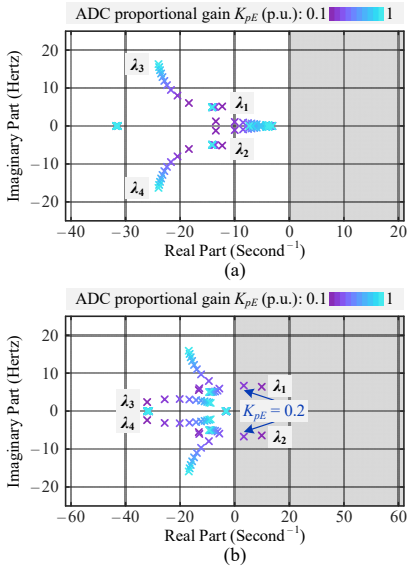

Fig. 12. Closed-loop pole trajectories of the SISO model with the proposed DAC method. (a) Weak grid conditions, SCR= 1.3. (b) Strong grid conditions, SCR= 6.7.

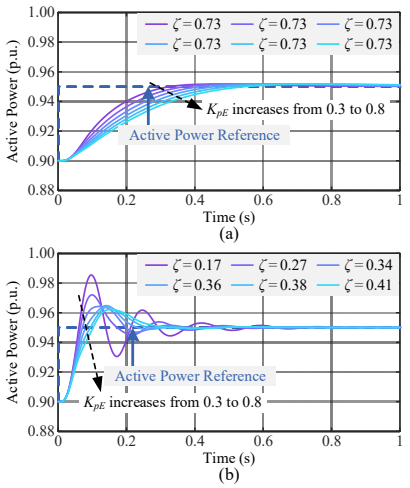

Fig. 13. Step response of the active power with the proposed DAC method. (a) Weak grid conditions, SCR= 1.3. (b) Strong grid conditions, SCR= 6.7.

$$V_{gq}^{ps} = -V_g \sin\delta \tag{34}$$

where V_g is the magnitude of the grid voltage.

Substituting (34) into (33) yields

$$E_q^{ps} = I_{gd}^{ps} X_g - V_g \sin\delta \tag{35}$$

If a grid disturbance results in $\tilde{P}>0$, $\tilde{\delta}>0$ according to the active power equation $P= EV_g\sin\delta/X_g$. Consequently, a positive $\tilde{\delta}$ initiates a negative $\tilde{E}_q^{qs}$ as derived in (35). Referring to the proposed control method in Fig. 10, $\tilde{\omega}_2<0$ is then generated, leading to $\tilde{\theta}<0$. The negative $\tilde{\theta}$ thus counteracts the variation in $\tilde{\delta}>0$ and $\tilde{P}>0$ caused by the grid disturbance, thereby enhancing system damping and alleviating the SSO issue.

Fig. 12 shows the closed-loop poles of the GFM IBR with the proposed DAC. The provided example focuses on the GFM IBR with the DSOGI-based SDM, characterized by a more compromised stability boundary as discussed in Section III. A K_P value of 0.05 p.u. is opted for illustration. Additionally, the integral gain K_{iE} is configured as $K_{iE}= K_{pE}\omega_E/10$ to make the proportional gain K_{pE} dominate the DAC dynamics [26]. Here, ω_E is exemplified as 30π.

It can be seen from Fig. 12(a) that the SSO mode poles $\lambda_{1\text{-}2}$ are located in the left-half plane (LHP) instead of the RHP as presented in Fig. 8(b), and $\lambda_{1\text{-}2}$ shift towards the LHP as K_{pE} increases. One additional pair of conjugate poles $\lambda_{3\text{-}4}$ can also be observed in Fig. 12(a), while they are located far from the RHP irrespective of the K_{pE} value. Therefore, the proposed DAC effectively mitigates the negative stability impact caused by the SDM under weak grid conditions. To achieve satisfactory system stability in a stiff grid, a K_{pE} value exceeding 0.2 p.u. is required, as shown in Fig. 12(b). With an increase in K_{pE}, $\lambda_{1\text{-}2}$ exhibit a substantial displacement away from the RHP, indicating enhanced stability. The addition of DAC also introduces an extra pair of conjugate poles $\lambda_{3\text{-}4}$ in Fig. 12(b). However, these poles are situated away from the RHP and thus exert negligible effects on system stability.

To opt for an appropriate K_{pE}, it is crucial to evaluate the damping ratio of the system response after a small disturbance. The step response of the active power, considering the proposed DAC method, is calculated using the closed-loop transfer function in (32), as shown in Fig. 13. Following the requirements specified in [15], a damping ratio (ζ) greater than 0.3 is essential. From Fig. 13. K_{pE} can be selected within the range of 0.5 p.u. to 0.8 p.u. to strike a trade-off between the rise time and overshoot. The stability margin for the viable K_{pE} values is then obtained using the open-loop transfer function in (32). Finally, $K_{pE}= 0.7$ p.u. is determined for its favorable stability margin in both weak grids (gain margin 28.5 dB and phase margin 70.5°) and stiff grids (gain margin 12 dB and phase margin 46.3°).

In the presence of distorted grid conditions, e.g., the PCC voltage comprises harmonics, the proposed control method remains applicable provided the VVC and VCC loops are configured with appropriate control architectures and parameters. The power control loops, such as APC and RPC, are normally designed to regulate the phase angle and amplitude of the fundamental-frequency PCC voltage, resulting in negligible interaction with voltage and current harmonics. Since the proposed control only alters the APC loop dynamics, it similarly has minimal impact on voltage and current harmonics. Assuming the droop gains of the APC and RPC loops are zero in Fig. 4, the proposed small-signal model can be simplified to derive the output admittance of the sequence-decomposed GFM IBR, thereby evaluating system stability under the influence of voltage harmonics. Addressing power quality issues caused by harmonics requires sophisticated harmonic suppression control schemes for the VVC and VCC loops, which is beyond the scope of this paper.

V. Controller Hardware-in-the-Loop Validation

CHIL tests are performed to validate the developed model and proposed control method presented in Sections III and IV. The CHIL test rig is based on OPAL-RT facilities, as shown in

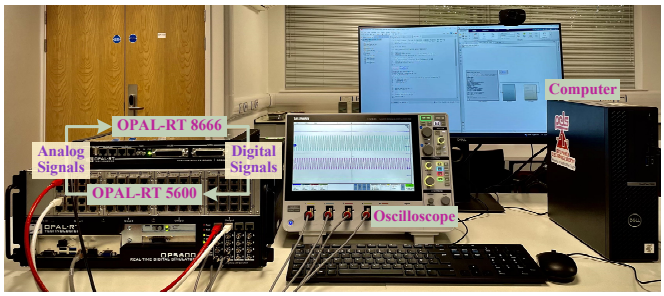

Fig. 14. Controller hardware-in-the-loop test rig.

Fig. 14. Specifically, the grid-connected system, including the physical components of the GFM IBR, LC filter, and AC grid, is emulated in the OPAL-RT 5600 real-time digital simulator. The control algorithms for the GFM IBR are implemented on the TMS320F28379D-based OPAL-RT 8666 DSP controller, with a sampling period of 50μs. During the experimental tests, the OPAL-RT 5600 outputs analog signals, such as PCC voltage, grid current, and filter current, to the OPAL-RT 8666. The DSP controller then outputs PWM digital signals back to the OPAL-RT 5600. The parameters used in the test are provided in TABLE I. As an illustration, the GFM IBR with the DSOGI-based SDM is employed.

Fig. 15 illustrates the SSO issue caused by the SDM under various grid conditions. When the APC droop gain is increased from 0.05 p.u. to 0.09 p.u. in a weak grid, divergent oscillations at a frequency of 4.9 Hz emerge. This confirms the calculation results of the developed model in Fig. 8(b), which predicts an unstable pole at 4.9 Hz when K_P= 0.09 p.u. Furthermore, Fig. 15(b) shows that, under strong grid conditions, merely raising K_P from 0.001 p.u. to 0.01 p.u. triggers a SSO phenomenon at 4.7 Hz, consistent with the results indicated by Fig. 9(b). These experimental findings align with the analysis results in Fig. 8 and Fig. 9, verifying the accuracy of the developed SISO model in Section III. The SSO observed in Fig. 15 highlights the challenge posed by integrating SDMs with GFM IBRs.

Fig. 16 shows the step response of the active power under different grid conditions. In the test, the proposed damping enhancement control is enabled with K_{pE}= 0.7 p.u. and K_P= 0.05 p.u. Comparing the results from the developed model using (32) in Fig. 13 with the experimental results in Fig. 16, the predictions of the developed model accurately capture the active power dynamics observed in the CHIL test. Thus, the model maintains high accuracy across different grid strengths. Additionally, Fig. 16 shows that the system remains stable under step changes in the active power reference, even with a large K_P value (0.05 p.u.) in a stiff grid. This corroborates the effectiveness of the proposed control method.

The experimental results evaluating the proposed DAC are further presented in Fig. 17, with K_{pE}= 0.7 p.u. The GFM IBR with SDMs stably operate under weak grid conditions, even when the APC droop gain is increased to 0.1 p.u. Similarly, Fig. 17(b) demonstrates the stable operation of the GFM IBR with step changes of the K_P value from 0.025 p.u. to 0.1 p.u. in a stiff grid. The above observations align with the analysis in Fig. 12 and validate the effectiveness of the DAC method in mitigating SSO issues.

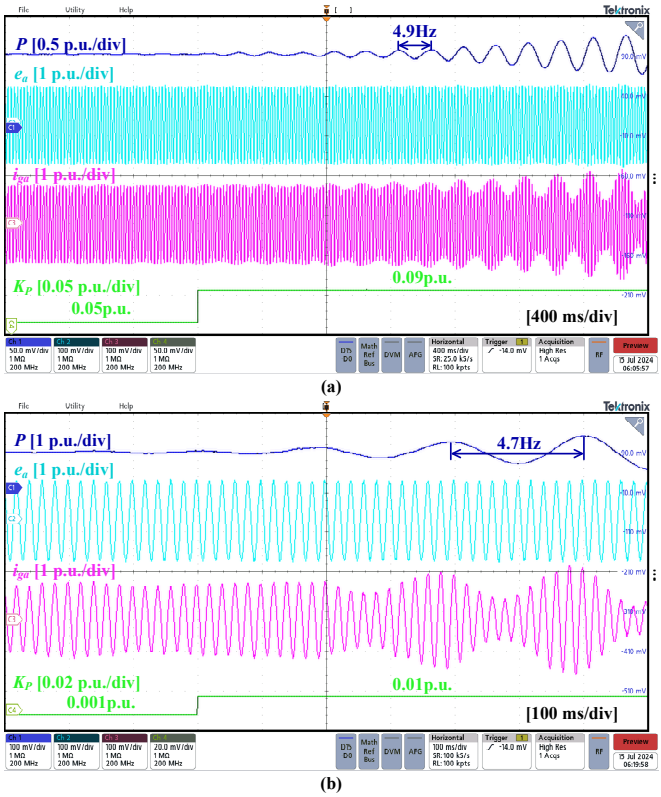

Fig. 15. CHIL test results under the step change of the APC droop gain. (a) Weak grid conditions, SCR= 1.3. (b) Strong grid conditions, SCR= 6.7.

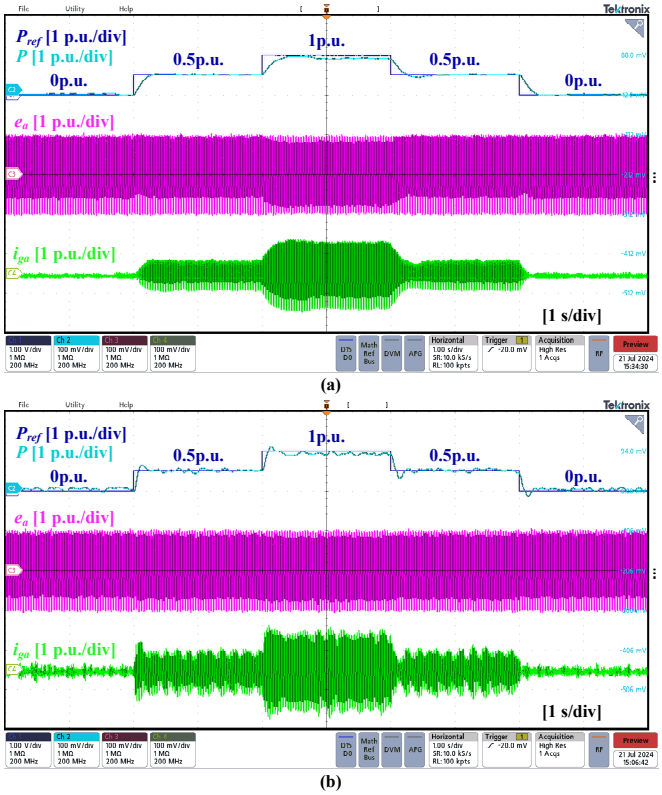

Fig. 16. CHIL test results under the step change of the active power reference considering the proposed DAC method. (a) Weak grid conditions, SCR= 1.3. (b) Strong grid conditions, SCR= 6.7.

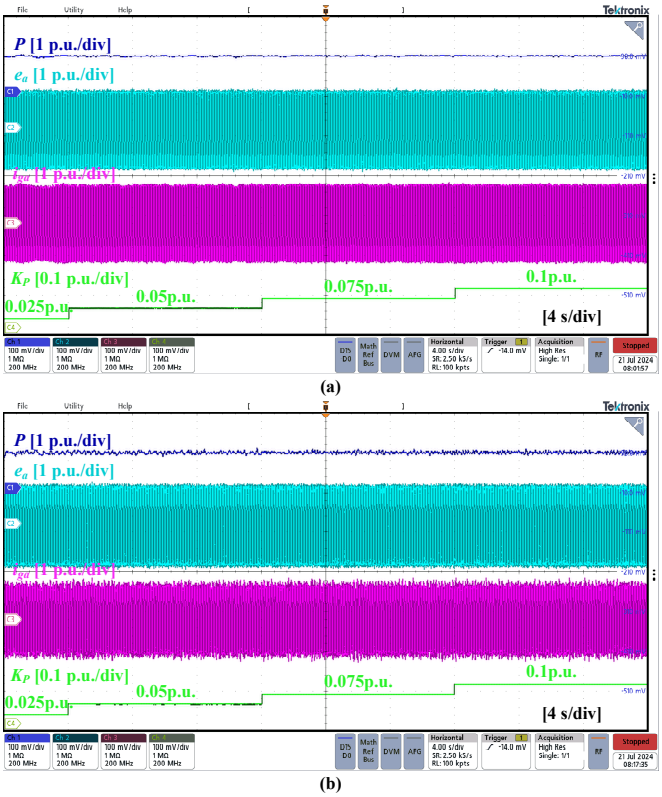

Fig. 17. CHIL test results under the step change of the APC droop gain considering the proposed DAC method. (a) Weak grid conditions, SCR= 1.3. (b) Strong grid conditions, SCR= 6.7.

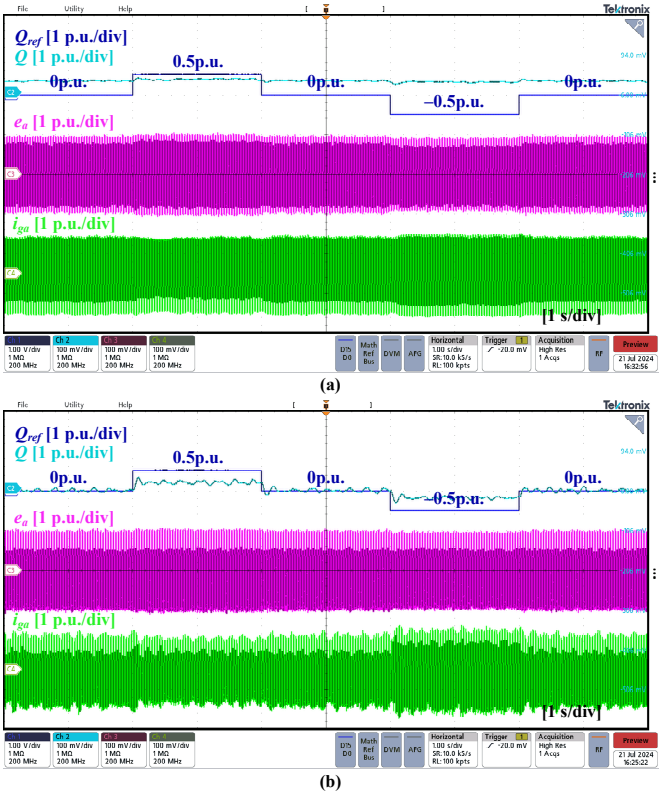

Fig. 18. CHIL test results under the step change of the reactive power reference considering the proposed DAC method. (a) Weak grid conditions, SCR= 1.3. (b) Strong grid conditions, SCR= 6.7.

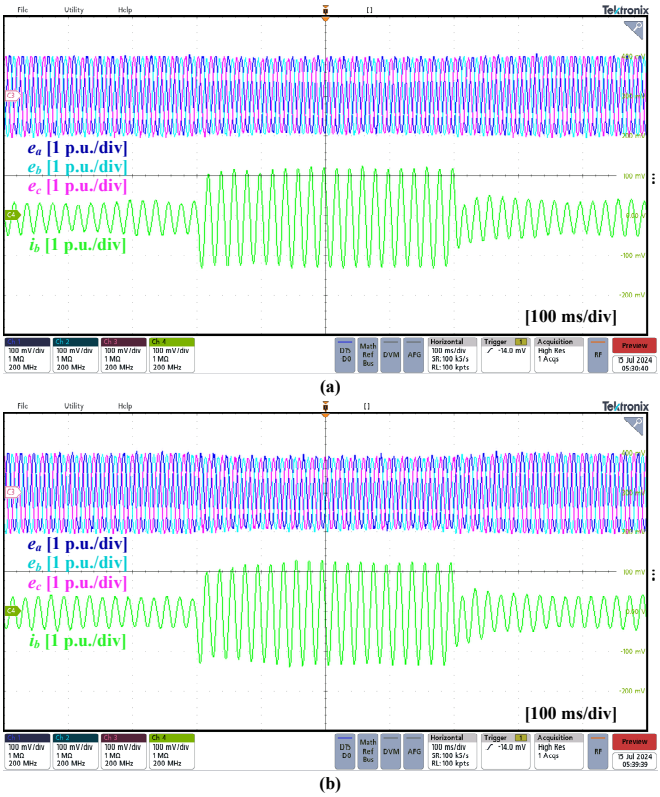

Fig. 19. CHIL test results under asymmetrical voltage sag conditions. (a) Single-phase voltage sag. (b) Two-phase voltage sag.

With the proposed control enabled (K_{pE}= 0.7 p.u. and K_P= 0.05 p.u.), step changes of the reactive power reference Q_{ref} are applied, and the results are shown in Fig. 18. By regulating Q_{ref} over a wide range of [–1 p.u., 1 p.u.], the system maintains stability, and the reactive power and PCC voltage follow the droop law under both weak and strong grid conditions. This substantiates the feasibility of proposed control under varying reactive power.

To demonstrate the effectiveness of the proposed method under asymmetrical voltage sag conditions, single-phase and two-phase grid voltage dips are performed during the test. Since this paper focuses on small-signal stability analysis, linearization is essential. Given this, the current limiter threshold is set to 1.5 p.u. and remains untriggered throughout the test to prevent nonlinearity. As detailed in Section III. B, the GFM IBR with SDMs is prone to unstable SSO issues, particularly with a large droop gain of APC. Therefore, a K_P value of 0.1 p.u. is considered. The test results, as shown in Fig. 19, feature P_{ref}= 0.4 p.u., Z_g= 0.3 p.u. (SCR= 3.3), and K_{pE}= 0.7 p.u. It can be found that the system consistently reaches the equilibrium point without instability issues. With the proposed control method, the PCC voltage maintains nearly nominal magnitude and relatively low degree of unbalance under both single-phase and two-phase voltage sag conditions. This aligns with the control objective of the sequence-decomposed GFM control. Hence, the proposed damping enhancement control successfully facilitates the stable operation of GFM IBRs with SDMs under asymmetrical voltage sag conditions.

VI. Conclusion

The SDM can result in cross-coupled dynamics, control delay, and control loop interactions in sequence-decomposed GFM IBRs. Under the same APC droop gain, GFM IBRs without SDMs can readily achieve stable operation, while those with SDMs are prone to SSOs. Based on the developed SISO small-signal model, the inherent instability mechanism in GFM IBRs with SDMs is studied. Comparative analysis reveals that GFM IBRs with the DSOGI-based SDM shows a poorer stability boundary than those with the DSC-based SDM. Taking the SDM into account, an enhanced hybrid synchronization controller is proposed to reshape the frequency response of GFM IBRs with SDMs. Test results show that the proposed method can significantly improve the stability robustness around the APC synchronization mode in a wide range of grid conditions. This study establishes a foundation for research on transient stability analysis and fault ride-through control design of GFM IBRs with sequence-decomposed control. Since this paper studies a typical two-area system, the developed SISO model cannot be directly applied when considering more than two oscillation areas. Further work is needed to develop a more practical model for a multi-area system, including the couplings within multiple sequence-decomposed GFM IBRs.

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Qian Liu received the B.S. and M.S. degrees in electrical engineering from North China Electric Power University, Beijing, China, in 2020 and 2023, respectively. He is currently a Marie Sklodowska-Curie Researcher and Ph.D. student at the School of Engineering, University of Warwick, Coventry, U.K. His research interests include modelling, stability analysis, and control design of the power electronics-dominated power systems.

Yanchang Liang received the B.S. and M.S. degrees from the School of Electrical Engineering at North China Electric Power University, China, in 2018 and 2021 respectively. He is currently a Marie Curie Early Stage Researcher and Ph.D. student at the University of Warwick, UK. His current research interests include control, optimization, reinforcement learning, parallel computing, and their applications in power systems.

Zuan Zhang received the B.S. degree in electronic information engineering from Zhejiang University, Hangzhou, China, in 2011, the M.S. degree in electrical engineering from Tsinghua University, Beijing, China, in 2014, and the Ph.D. degree in engineering from the University of Warwick, Coventry, UK, in 2024. He is currently a Senior Engineer with Siemens Energy, Manchester, UK. He was a Research Engineer and Project Manager with the Electric Power Research Institute, China Southern Power Grid, Guangzhou, China, from 2014 to 2020. He was a Marie Curie Early-Stage Researcher at the University of Warwick, from 2020 to 2023. His research interests include wind energy systems, HVDC, and intelligent control.

Miao Wang received the B.S. degree in electrical engineering from North China Electric Power University, Beijing, China, in 2020, and the M.S. degree in electrical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2023. She is currently working toward the Ph.D. degree in electrical engineering with the School of Engineering, University of Warwick, Coventry, U.K. Her current research interests include, wind farm control, wind turbine control and reinforcement learning.

Xiaowei Zhao received the Ph.D. degree in control theory from Imperial College London, London, U.K., in 2010. He was a Post-Doctoral Researcher with the University of Oxford, Oxford, U.K., for three years before joining the University of Warwick, Coventry, U.K., in 2013. He is currently Professor of control engineering with the School of Engineering, University of Warwick. His main research areas are control theory and machine learning with applications in offshore renewable energy systems, smart grids, and autonomous systems.